Math 141, F11 **Name:** Answer
Exam 1 **Section:** **SSN:** xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	18	
3	24	
4	22	
5	16	
Total	100	

Good Luck !

Problem 1. (20 points) Let $f(x) = -x^2 + 4x$ and $g(x) = \sqrt{x}$.

(a) Find a formula for the composite function $f \circ g$.

$$f \circ g(x) = f(g(x))$$
$$= -(\sqrt{x})^2 + 4\sqrt{x}$$
$$= -x + 4\sqrt{x}$$

(b) What is the domain of $f \circ g$?

$x \geqslant 0$ or $[0, +\infty)$

(c) Find a formula for the composite function $g \circ f$.

$$g \circ f(x) = g(f(x))$$
$$= \sqrt{-x^2+4x}$$

(d) What is the domain of $g \circ f$?

$$-x^2+4x \geqslant 0$$
$$x^2-4x \leqslant 0$$
$$x(x-4) \leqslant 0$$
$$\Rightarrow 0 \leqslant x \leqslant 4 \quad \text{or} \quad [0,4]$$

Problem 2. (18 points)

(a) Find a formula for the inverse of the function $f(x) = \dfrac{3x-1}{2x+3}$.

write $y = \dfrac{3x-1}{2x+3}$

$\Rightarrow 2xy + 3y = 3x - 1$

$2xy - 3x = -1 - 3y$

$x(2y-3) = -1-3y$

$x = \dfrac{-1-3y}{2y-3} \quad \Rightarrow f^{-1}(x) = \dfrac{-1-3x}{2x-3}$

(b) Simplify the following expression

$$3^3 \cdot 3^{-2} + \log_4 2 + \log_4 8 + e^{\frac{3}{2}\ln 4}$$

$= 3^{3-2} + \log_4(2\cdot 8) + e^{\ln(4)^{\frac{3}{2}}}$

$= 3 + \log_4 16 + 4^{\frac{3}{2}}$

$= 3+2+8$

$= 13$

(c) Determine if the following function is even, odd, or neither:

$$f(x) = x^2 \sin x$$

$f(-x) = (-x)^2 \sin(-x)$

$= x^2(-\sin x)$

$= -x^2 \sin x$

$= -f(x) \Rightarrow f(x)$ is odd.

Problem 3. (24 points) Find each of the following limits if it exits.

(a) $\lim_{x\to 1} e^{x^3-x}$

$$= e^{1^3-1} = e^0 = 1$$

(b) $\lim_{x\to -2} \dfrac{x^2-4}{x^2-x-6}$

$$= \lim_{x\to -2} \frac{(x-2)\cancel{(x+2)}}{(x-3)\cancel{(x+2)}}$$

$$= \frac{-2-2}{-2-3} = \frac{4}{5}$$

(c) $\lim_{x\to +\infty} \dfrac{x^3-1}{2x^3+x-10}$

$$= \lim_{x\to +\infty} \frac{\frac{x^3-1}{x^3}}{\frac{2x^3+x-10}{x^3}} = \lim_{x\to +\infty} \frac{1-\frac{1}{x^3}}{2+\frac{1}{x^2}-\frac{10}{x^3}}$$

$$= \frac{1}{2}$$

(d) $\lim_{x\to \frac{\pi}{2}^+} \dfrac{1}{\cos x} = -\infty$

$\downarrow$

$\cos x \to 0^-$ as $x \to \frac{\pi}{2}^+$

Problem 4. (22 points)
(a) Explain why this function is discontinuous at $x = 0$:

$$f(x) = \begin{cases} \dfrac{x^2 - 3x}{|x|} & x \neq 0, \\ -3 & x = 0. \end{cases}$$

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{x^2-3x}{|x|} = \lim_{x\to 0^-} \frac{x(x-3)}{(-x)}$$

$$= \lim_{x\to 0^-} -(x-3) = 3 \neq f(0)$$

$\Rightarrow f(x)$ is discontinuous at $x=0$.

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{x^2-3x}{x} = \lim_{x\to 0^+} \frac{x(x-3)}{x}$$

$$= \lim_{x\to 0^+} x-3 = -3$$

(b) Let $f(x) = x^3 + x$. Find $f'(x)$ and $f''(x)$ using the definitions.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{[(x+h)^3+(x+h)]-[x^3+x]}{h}$$

$$= \lim_{h\to 0} \frac{[x^3+3x^2h+3xh^2+h^3+x+h]-[x^3+x]}{h}$$

$$= \lim_{h\to 0} \frac{3x^2h+3xh^2+h^3+h}{h}$$

$$= \lim_{h\to 0} 3x^2+3xh+h^2+1 = 3x^2+1.$$

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h)-f'(x)}{h} = \lim_{h\to 0} \frac{[3(x+h)^2+1]-[3x^2+1]}{h}$$

$$= \lim_{h\to 0} \frac{[3x^2+6xh+3h^2+1]-[3x^2+1]}{h}$$

$$= \lim_{h\to 0} \frac{6xh+3h^2}{h}$$

$$= \lim_{h\to 0} 6x+3h = 6x.$$

Problem 5. (16 points) Let $f(x) = \dfrac{1}{2x-1}$.

(a) Find $f'(x)$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{2(x+h)-1} - \frac{1}{2x-1}}{h} = \lim_{h \to 0} \frac{\frac{1}{2x+2h-1} - \frac{1}{2x-1}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{(2x-1)-(2x+2h-1)}{(2x+2h-1)(2x-1)}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{-2h}{(2x+2h-1)(2x-1)}}{h}$$

$$= \lim_{h \to 0} \frac{-2}{(2x+2h-1)(2x-1)}$$

$$= -\frac{2}{(2x-1)^2}$$

(b) Use the result of (a) to find an equation of the tangent line to the curve $y = \dfrac{1}{2x-1}$ at the point $(1, 1)$.

$$m = f'(1) = -\frac{2}{(2 \cdot 1 - 1)^2} = -2$$

$$\frac{y-1}{x-1} = f'(1)$$

$$\Rightarrow \frac{y-1}{x-1} = -2$$

$$y - 1 = -2x + 2$$

$$y = -2x + 3$$